IB MATHEMATICS SL • UNIT 4.01

C Topic 4: Statistics, Regression & Probability Distributions

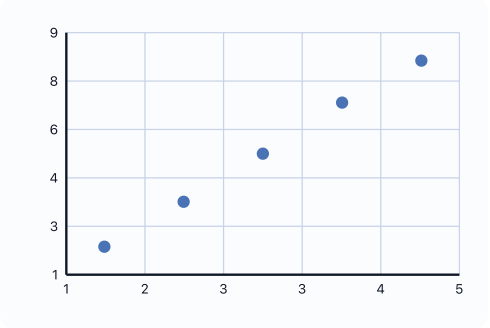
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Topic 4: Statistics, Regression & Probability Distributions Practice Exercises

1 MCQ

If $P(A) = 0.3$ and $P(B) = 0.5$ and A, B are independent, find $P(A \cup B)$.



A 0.65

B 0.80

C 0.15

D 0.85

2 MCQ

The following bar chart shows the number of students who achieved different grades in a mathematics test.\nThe grades are A, B, C, and D. The data is as follows: Grade A: 14 students, Grade B: 22 students, Grade C: 18 students, Grade D: 26 students.\nWhat is the total number of students who took the test?

A 70

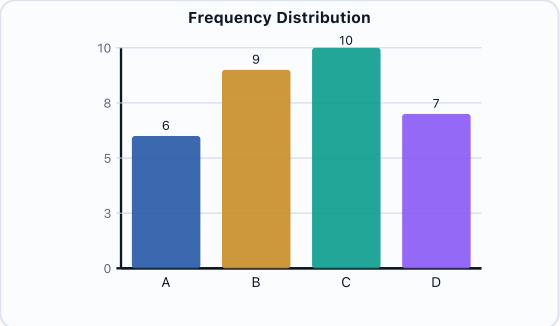
B 80

C 78

D 60

3 MCQ

The bar chart below shows the number of students who chose each of four different sports as their favorite. What is the total number of students surveyed?



A 50

B 60

C 70

D 80

4 MCQ

The histogram below shows the number of hours 100 students spent studying for an exam. The number of students who studied for between 0 and 5 hours is represented by a bar with a height of 30. Determine the number of students who studied for between 5 and 10 hours.

A 20

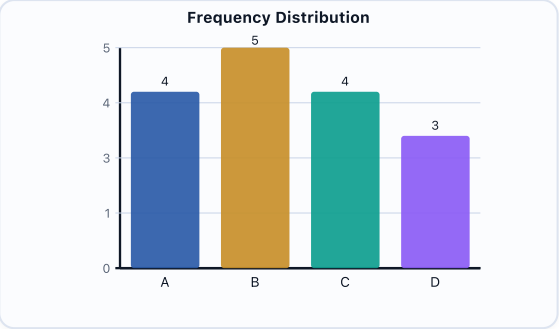
B 25

C 30

D 35

5 MCQ

The bar chart below shows the number of students who achieved a certain score on a test. What is the total number of students who scored 6 or 8?



A 30

B 28

C 26

D 32

6 MCQ

The following two-way table shows the results of a survey on favourite sports among a group of students. Calculate the total number of students surveyed.

A 50

B 45

C 60

D 55

7 MCQ

Find the mean of the data set: $20, 13, 12, 12, 8$.

A 10

B 13

C 8

D 11

8 MCQ

Find the mean of the data set: $10, 1, 5, 14, 3, 7, 0, 8$.

A 6

B 4

C 3

D 1

9 MCQ

Find the mean of the data set: $31, 3, 13, 4, 5, 4$.

A 6

B 5

C 10

D 7

10 MCQ

Find the mean of the data set: $-2, 6, 7, 8, 11, 6$.

A 4

B 3

C 1

D 6

11 MCQ

Calculate the mean of the data set: $17, 20, 19, 25, 14$.

A 16

B 19

C 14

D 15

12 MCQ

Find the mean of the data set: $12, -10, 1, 7, 14, 3, 9, 12$.

A 6

B 4

C 1

D 3

13 MCQ

A bag contains 4 red balls, 3 green balls, and 4 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

A $\frac{8}{11}$

B $\frac{2}{11}$

C $\frac{3}{11}$

D $\frac{4}{11}$

14 MCQ

A bag contains 4 green balls, 3 blue balls, and 2 red balls. A ball is drawn at random. Find $P(\text{blue})$.

A $\frac{4}{9}$

B $\frac{1}{3}$

C $\frac{2}{9}$

D $\frac{2}{3}$

15 MCQ

A bag contains 8 green balls, 6 blue balls, and 6 red balls. A ball is drawn at random. Find $P(\text{red})$.

- **A** $\frac{7}{20}$
- **B** $\frac{2}{5}$
- **C** $\frac{3}{10}$
- **D** $\frac{7}{10}$

16 MCQ

Find the mean of the data set: $2, 14, -3, 0, 7$.

- **A** 1
- **B** 2
- **C** 4
- **D** 0

17 MCQ

A bag contains 3 green balls, 7 red balls, and 4 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

- **A** $\frac{2}{7}$
- **B** $\frac{1}{2}$
- **C** $\frac{11}{14}$
- **D** $\frac{3}{14}$

18 MCQ

Calculate the mean of the data set: $11, 17, 22, 27, 12, 25$.

- **A** 14
- **B** 16
- **C** 19
- **D** 15

19 MCQ

A bag contains 4 red balls, 2 blue balls, and 2 yellow balls. A ball is drawn at random. Find $P(\text{blue})$.

- **A** $\frac{1}{2}$
- **B** $\frac{1}{4}$
- **C** $\frac{3}{8}$
- **D** $\frac{3}{4}$

20 MCQ

A bag contains 4 yellow balls, 6 blue balls, and 3 green balls. A ball is drawn at random. Find $P(\text{green})$.

- **A** $\frac{6}{13}$
- **B** $\frac{3}{13}$
- **C** $\frac{4}{13}$
- **D** $\frac{10}{13}$

21 MCQ

A bag contains 5 green balls, 5 red balls, and 3 yellow balls. A ball is drawn at random. Find $P(\text{yellow})$.

A $\frac{4}{13}$

B $\frac{3}{13}$

C $\frac{5}{13}$

D $\frac{10}{13}$

22 MCQ

Find the mean of the data set: $9, 9, -3, 4, 1, 11, 12, -3$.

A 5

B 3

C 1

D 2

23 MCQ

Compute the mean of the data set: $12, 9, 13, 12, 29$.

A 10

B 12

C 15

D 11

24 MCQ

A bag contains 4 green balls, 4 red balls, and 5 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

A $\frac{5}{13}$

B $\frac{4}{13}$

C $\frac{3}{13}$

D $\frac{9}{13}$

25 MCQ

Calculate the mean of the data set: $22, 26, 15, 26, 16, 20, 13, 14$.

A 14

B 15

C 16

D 19